

Changeover

Chief Economist • Halvern Currency Board

15 DAYS • 49 STOPS • GRADE 12 • CHANGEOVER

THE OPENING CARD

In fourteen days the shops and banks of Halvern stop taking the mark, and no one has said what the new money is worth. You are the chief economist to the currency board, and the rate goes out with your name on it. In fifteen days you hand over the Halvern rate book, and the rate is its last page. Nine hundred thousand people are paid in marks. Set it too high and the reserves go in a week; set it too low and every wage buys less bread.

Fourteen days and no rate

BEFORE THE DAY — GREET

Get round the four floors before the counter opens

Ngozi Okonjo, the counter supervisor, wants you known to as many of the people on these four floors as you can get round this morning.

PLAN CARD 129W

Monday, fourteen days out. Ada Verhoeven, the board chair, wants a rate on Friday. Wages and contracts have to be written against something. Emil Radic, the chief statistician, has been handed a basket of goods fixed years ago. Half of Halvern heated with coal back then. He will not certify an index built on it. Ngozi Okonjo, the counter supervisor, has forty people at her window. Mira Vasek is at the front of it with a tin of notes and one question about Monday. Today you settle what the board actually measures. Page one of the rate book is due tonight: the counted statistics, each with the period it counts beside it. A figure the board cannot defend on the fifteenth comes back in six months as a fact.

BRIEFING 14W

Three weeks of prices, a basket from another decade, and a rate due Friday.

WHAT YOU DECIDE 10W

Establish what the board measures and what it can defend.

1.1 The same list, twice

PRICE ROOM · A ROOM · BALLPARK

SCENE 28W

Radic has the fixed basket priced twice. It cost 84 marks in the base year and 231 marks this month. The board wants an index number for Friday.

GUIDE 55W

An index is a ratio dressed as a number. Divide what the basket costs now by what the same basket cost in the base year, then multiply by a hundred so the base year reads as 100. Work it out, and note what the result does and does not say about anybody's cost of living.

ASK 7W

Estimate the price index for this month.

BEHIND THE BUTTON 89W

Why the basket is held fixed. If the list changed with the prices, a rise could be hidden by substituting something cheaper, and the index would measure the substitution rather than the prices. Holding it still is what makes two years comparable.

What holding it still costs. A basket fixed years ago prices a city that has changed — which is Radic's objection. Both facts are true at once: the basket has to be fixed to be an index, and it has to be revised to be about anybody.

TAKES AS READ

the basket is the same list of goods in both years · gross domestic product, and the four things it adds up — taken as read

VERDICT 110W

Two hundred and thirty-one over eighty-four is 2.75, and multiplying by a hundred gives an index of 275. The base year reads 100 by construction, which is the only reason index numbers are comparable at all. What this is not is an inflation rate: 275 is a level, and a rate is what the level does between two dates. Reading the difference of 147 as the answer gives a number in marks rather than in index points, and it cannot be compared with any other month. The basket is the part Radic is right about, and it does not change this arithmetic — it changes what the arithmetic is about.

TAKEAWAY 13W

An index is a ratio to a base year, multiplied by a hundred.

1.2 Forty people, and what they have heard

COUNTER ROOM • A PERSON • CHOICE

SCENE 22W

Okonjo's queue was eleven people on Friday and forty this morning. Nothing has been announced. The board has not met since Thursday.

GUIDE 41W

A queue is a measurement of what people expect rather than of what has happened. Ask what would make somebody move today rather than next week, and which of the four explanations on offer is consistent with nothing having been announced.

ASK 7W

What is the lengthening queue evidence of?

BEHIND THE BUTTON 84W

Why expectations are a policy variable rather than a mood. People act on what they think the rate will be, so the belief changes behaviour before any policy does. A rumour is therefore a monetary event, and the board's silence is a statement whether it means to be or not.

What the counter measures that the index cannot. The index reports prices already charged. The queue reports what people think prices will do, which is the variable the board can actually move on Friday.

TAKES AS READ

nothing has been announced since Thursday • the short-run trade-off between inflation and unemployment — taken as read

VERDICT 101W

Nothing has been announced, so nothing has happened to prices or to the rate — and forty people have still decided that today is better than next week. That decision is about the future rather than the past, which makes the queue a measurement of expectations and the only one the board has. Weekend prices would show in Radic's figures rather than at the counter. Staffing changes how fast a queue moves, not how many people join it. And the old currency is nowhere near exhausted: Bram Tulloch, who manages the note room, has weighed a fraction of what was issued.

TAKEAWAY 12W

Expectations move before policy does, and a silence is read as information.

1.3 Money on a platform scale

NOTE ROOM • A ROOM • VERIFY

SCENE 31W

Tulloch has 214 kilograms of returned notes on the platform scale. A sample says the mix averages 0.94 grams a note. The board needs a count for the money supply figure.

GUIDE 51W

Counting this many notes by hand is not possible, so the count comes from a mass and a mass per note. Divide, mind the units, and then ask what the answer depends on that is not on the scale. The sample is doing as much work here as the weighing is.

ASK 13W

Predict the count from the weight, then hand-count a bundle and check it.

BEHIND THE BUTTON 88W

What the sample decides. Notes of different denominations weigh nearly the same, so the mass gives a count of notes and says nothing about their value. The denomination mix comes from a sample, and the value of the return is the count multiplied by that mix.

Why the count matters to the board. The money supply figure is what the rate is argued against. The part of it that is cash is exactly this: what was issued, less what has come back and been burnt under two signatures.

TAKES AS READ

the sampled mass per note is representative of the whole return • demand, supply and the price that clears a market — taken as read

VERDICT 114W

Two hundred and fourteen kilograms at 0.94 grams a note is about 228,000 notes. One bundle hand-counted says the notes average 0.98 grams, so the true count is nearer 218,000. That is four per cent fewer, and the error is in the sample rather than in the scale. That matters twice over. The count feeds the money supply figure. The same sample supplies the denomination mix that turns a count into a value. So a mix wrong by four per cent is a value wrong by about the same. Notes of different denominations weigh almost the same, which is why the scale can give a count and can never give an amount of money.

TAKEAWAY 17W

A weighed count is a count of notes, and their value needs a sampled mix as well.

END OF THE DAY 10W

Every economic statistic counts something specific, over a specific period.

What Halvern actually makes

PLAN CARD 107W

Tuesday. The board has an output figure it cannot defend. Radic found the flour counted twice. It went in once as a sale to the bakery. Then it went in again inside the bread. Della Prentice, the labour statistician, has a problem of her own. Her unemployment rate fell in a month when three factories shut. Verhoeven needs both numbers before the rate can be argued about at all. Today you work out what the output figure should really be. You also say what the labour figure is actually counting. Two statistics that move the wrong way are usually measuring something other than what their names say.

BRIEFING 10W

The board's own output figure has flour in it twice.

WHAT YOU DECIDE 12W

Add up the output and count who is working, without double-counting either.

2.1 Flour, and then bread

PRICE ROOM · A ROOM · BALLPARK

SCENE 27W

The month's figures: households spent 412 million, firms invested 96, the government spent 178, exports were 64 and imports 121. The board's draft adds the imports in.

GUIDE 47W

Add what was spent on final goods by each of the four buyers, and subtract imports rather than adding them. An import is spending that went to somebody else's output. Work out the figure the board should publish, then say what the draft's error does to it.

ASK 5W

Estimate the month's output figure.

BEHIND THE BUTTON 79W

Why imports are subtracted. Household spending already includes what households bought from abroad, so leaving imports in counts foreign output as Halvern's. Subtracting them removes exactly that, which is why the term is exports less imports rather than exports.

Why the flour is a different error. Counting flour sold to a bakery and then the bread counts the same wheat twice. Only final goods go in, which is why intermediate sales are excluded rather than being a separate component.

TAKES AS READ

the four spending figures are all for the same month · demand, supply and the price that clears a market — taken as read

VERDICT 99W

The four components add to 750 and imports of 121 come off, leaving 629 million. Adding imports instead of subtracting them is not a small slip. It moves the figure by twice the import total, 242 million, because the error is in the direction as well as the term. The reason imports are subtracted is that household spending already contains them, so leaving them in counts another country's output as Halvern's. The flour is a separate error of the same family: an intermediate good counted once as flour and again as bread. It is excluded rather than netted off.

TAKEAWAY 11W

Imports are subtracted because they are spending on somebody else's output.

2.2 A rate that fell in a bad month

COUNTER ROOM • A PERSON • BALLPARK

SCENE 23W

Prentice has 41,000 unemployed out of a labour force of 486,000, down from 44,000 out of 502,000. Three factories shut during the month.

GUIDE 37W

Work out both rates and compare them. Then look at what happened to the denominator. The labour force fell by sixteen thousand in a month when three factories closed, and that is the whole of the story.

ASK 5W

Estimate this month's unemployment rate.

BEHIND THE BUTTON 86W

Who is in the labour force. People working, and people not working who are looking for work. Somebody who stops looking is in neither category and leaves the calculation entirely, which lowers the numerator and the denominator at once.

Why a falling rate can be bad news. If people leave the labour force faster than they find work, the rate falls while fewer people are working. The figure that does not have this problem is the participation rate, which is why it is published beside it.

TAKES AS READ

both months were counted the same way • gross domestic product, and the four things it adds up — taken as read

VERDICT 92W

Forty-one thousand over 486,000 is 8.4 per cent, against 8.8 last month, so the published rate improves. The labour force fell by sixteen thousand in the same month, and that is where the improvement came from. People who stop looking are in neither the numerator nor the denominator. So they leave the rate rather than raising it. Three factories closing and a rate falling are consistent, and that is the point. Using last month's labour force with this month's unemployed gives 8.2, which mixes two months and makes the fall look larger.

TAKEAWAY 14W

A rate can fall because its denominator shrank rather than because its numerator did.

2.3 Somebody else's loan

LEDGER HALL • A ROOM • CHOICE

SCENE 38W

Sunita Iyer, the banking supervisor, has the eleven regional balance sheets. Between them they hold 62 million marks of reserves against 2.4 billion of deposits. Ruth Haldane, who manages the largest of the eleven, is in the room.

GUIDE 39W

Look at the two numbers and say what they mean about what a bank is. Then decide which of the four statements about the deposits is true, because three of them describe an arrangement no bank has ever had.

ASK 9W

What is true of the 2.4 billion in deposits?

BEHIND THE BUTTON 90W

Why holding a fraction is normal rather than a scandal. Depositors do not all arrive on the same morning, so a bank can lend most of what it holds and still meet ordinary demands. The arrangement works on the strength of that assumption and nothing else.

What that makes the money supply. Not a quantity somebody decided on: the deposits are money, and they exist because loans were made. So the money supply is largely something the banking system created, which is why a reserve requirement is a policy tool.

TAKES AS READ

the eleven balance sheets are for the same date • what counts as money, and the three things it has to do — taken as read

VERDICT 102W

Sixty-two million of reserves stands behind 2.4 billion of deposits, which tells you at once that the deposits are not cash in a vault. There is not remotely enough cash for that. They exist because banks lent, and a loan credited to somebody's account is a deposit. That is what makes the money supply something the banking system created rather than something the board issued. It is why deposits can exceed issued currency by a wide margin. The board's reserves do not guarantee them either; sixty-two million against 2.4 billion would be a guarantee of about two and a half per cent.

TAKEAWAY 12W

Most of what a country calls money was never issued by anybody.

END OF THE DAY 15W

An aggregate is a sum of specific things, and what it excludes is a decision.

A level and a rate are different things

PLAN CARD 97W

Wednesday. The board's draft statement uses the word inflation for the index level. Those are two different things. Radic will not have it published that way. Douglas Farr, the press officer, agrees for his own reason. A correction costs more than the sentence saves. Meanwhile the tram drivers have a wage claim written in marks. Nobody has said what those marks buy. Today you turn the index level into a rate, and the wage into real terms. A level quoted as a rate is the commonest error in a money statement. It always points the alarming way.

BRIEFING 10W

The board keeps quoting the index and calling it inflation.

WHAT YOU DECIDE 15W

Turn two index numbers into an inflation rate, and a wage into what it buys.

3.1 From 248 to 275

PRICE ROOM · A ROOM · BALLPARK

SCENE 19W

The index was 248 last month and 275 this month. The board's draft says inflation is 275 per cent.

GUIDE 47W

A rate is a change divided by where it started. Take the difference between the two index values, divide by the earlier one and express it as a percentage. Then compare that with what the draft says, because the two numbers differ by a factor of twenty-five.

ASK 5W

Estimate the month's inflation rate.

BEHIND THE BUTTON 87W

Why the confusion is so common. The index is a percentage-looking number and the rate is a percentage, so both read as per cent and only one of them is. An index of 275 says the price level is 2.75 times the base year; it says nothing about this month.

Why the direction of the error matters. Quoting a level as a rate always overstates inflation, and an overstated inflation figure raises exactly the expectations the board is trying to hold down. The error is not neutral.

TAKES AS READ

both index values are on the same base year · a price index, and the basket it is an index of — taken as read

VERDICT 105W

The change is 27 index points on a base of 248, which is 10.9 per cent for the month. The draft's 275 is the index level, and quoting it as a rate overstates inflation by a factor of twenty-five. The error always runs that way, because a level is a much bigger number than the rate that produced it. And the direction is what raises the expectations the board is trying to hold down. Eleven per cent a month is the honest figure, and it is not comforting. Compounded it is about 250 per cent a year, which is what the statement should say instead.

TAKEAWAY 18W

A rate is a change over where it started, and an index is not a change at all.

3.2 The same wage, deflated

COUNTER ROOM • A PERSON • BALLPARK

SCENE 24W

The tram drivers earned 890 marks a month in the base year and 2,180 now. The index is 275. Their claim asks for 2,600.

GUIDE 50W

A wage in the money of the day cannot be compared with one from another year until the price change is taken out. Divide the current wage by the index over a hundred, and set the answer against the base-year figure. Then say what the claim is actually asking for.

ASK 8W

Estimate the drivers' current wage in base-year terms.

BEHIND THE BUTTON 96W

What deflating does. Dividing by the price level converts money-of-the-day into money of the base year, so two figures can be compared by what they would buy. It is the same operation whether the figure is a wage, an output total or a tax take.

Why the claim reads differently once deflated. In base-year terms the drivers are on 793 marks against 890 before, so they have lost about eleven per cent. The claim for 2,600 is 945 in real terms — six per cent above where they started rather than a 19 per cent rise.

TAKES AS READ

the index applies to what the drivers actually buy • inflation as a rate, and the difference between a level and a rate — taken as read

VERDICT 91W

Dividing 2,180 by the price level of 2.75 gives 793 base-year marks, against 890 in the base year. The drivers are about eleven per cent worse off, and their nominal wage has more than doubled. That is the whole reason the distinction exists. It is also what makes the wage claim arguable rather than greedy. In real terms 2,600 is 945, six per cent above where they started. Multiplying by the index instead of dividing gives 5,995, which is the error that makes any historical comparison absurd in the flattering direction.

TAKEAWAY 14W

Two wages from different years compare only after the price change is divided out.

3.3 Which side it came from

RATE ROOM • A ROOM • CHOICE

SCENE 30W

Prices are up eleven per cent for the month. Output is down four per cent and unemployment is up. The treasury liaison, Josiah Mbeya, has a draft blaming excess demand.

GUIDE 38W

Two of these three numbers move together under one explanation and against each other under the other. Work out which pattern demand-driven inflation produces and which a cost shock produces, then say which the figures are consistent with.

ASK 10W

Which explanation fits prices up, output down and unemployment up?

BEHIND THE BUTTON 71W

What demand-pull looks like in the figures. Prices and output rise together, because more spending pulls both up, and unemployment falls. That is the pattern the draft assumes.

What a cost shock looks like. Prices rise while output falls, because the same production costs more, and unemployment rises with it. Halvern imports its flour and fuel and the exchange rate has moved, which is a textbook cost shock arriving by wire.

TAKES AS READ

the output and price figures are for the same month • short-run aggregate supply, and what a cost shock does to it — taken as read • inflation as a rate, and the difference between a level and a rate — taken as read

VERDICT 103W

Demand-driven inflation raises prices and output together and lowers unemployment, because more spending pulls on both. Here output is falling and unemployment rising while prices rise. That is the opposite pattern, and the signature of costs. The same production is more expensive, so less of it happens at a higher price. Halvern imports flour and fuel and the exchange rate has moved, so the mechanism is on the wires rather than in the shops. A rise in capacity would lower prices, and a fall in the money supply would lower them too. Both are the wrong direction on the one number nobody disputes.

TAKEAWAY 11W

Prices and output moving together is demand; moving apart is cost.

3.4 Money on a platform scale — Review

NOTE ROOM • A ROOM • CALLBACK • PROTOCOL

SCENE 41W

Femi Adeyinka, the vault officer, has four things in front of him that Halvern is using as money this fortnight. Only one of them was issued by the board. Tulloch wants each one placed against the jobs money has to do.

GUIDE 55W

Money has three jobs: it settles a payment, it holds value between now and later, and it is the ruler prices are quoted in. Take each of the four in turn and ask which job it is doing and which it is failing. Nothing here fails all three, and one of them is failing none.

ASK 13W

Match each thing to the job of money it is doing or failing.

BEHIND THE BUTTON 91W

Why the three jobs are listed separately. Most of the time one thing does all three and nobody has cause to tell them apart. A changeover pulls them apart for a fortnight, which is why a currency board is the one place the distinction is operational rather than academic.

Which job fails first under inflation. Holding value goes before settling payments does. People keep taking the notes and stop keeping them. So the same money changes hands faster while nobody wants to be holding it at the end of the week.

TAKES AS READ

all four are circulating in Halvern during the changeover fortnight • demand, supply and the price that clears a market — taken as read

VERDICT 87W

The three jobs come apart during a changeover, which is the only time most people notice there are three. Old marks still settle payments, because acceptance is all that job asks and the counter is still open. Scrip half the city will not take fails there, whatever is printed on it. Wages held back are a store-of-value failure and nothing else; they would still buy bread today. And two currencies chalked side by side is one of them working purely as the unit prices are quoted in.

TAKEAWAY 16W

Money does three separate jobs, and a thing can do one of them while failing another.

END OF THE DAY 14W

An index is a level; inflation is what the level did between two dates.

The city the basket prices

BEFORE THE DAY — FOLLOW

Follow Novotny round her repricing

Kata Novotny, a market trader from Vend Street, brings her price cards up to the price room every morning and does not stop walking to explain them. Stay with her: the basket Emil Radic, the chief statistician, has been handed prices a city that stopped existing, and her stall — visible from the window, four hundred metres west — is the only place in Halvern where the new one can be checked against something real.

PLAN CARD 107W

Thursday. Radic has refused to certify for the second time. This is the day he is right about. The basket was fixed when half of Halvern heated with coal. Nobody bought imported flour then. So it prices a city that has stopped existing. Verhoeven needs a certified figure to set a rate against on Friday. Kata Novotny has three weeks of stall prices. They are the only real check on any new basket. Today you decide what the basket should hold, and what changing it costs. Fixing a basket is what makes an index comparable over time. Refusing to ever revise it is what makes it wrong.

BRIEFING 10W

Radic will not sign, and he is right about why.

WHAT YOU DECIDE 11W

Fix the basket before the index built on it is certified.

4.1 Coal, and imported flour

PRICE ROOM · A ROOM · TRIAGE

SCENE 27W

The current basket gives coal a fifth of its weight and imported flour nothing. Novotny's stall and the counter queue both say the opposite is now true.

GUIDE 42W

A basket is a set of weights, and the weights are supposed to be what households actually spend on. Work out what the present weights get wrong, and pick the change that fixes the index rather than the change that flatters it.

ASK 9W

What should be done about the basket before Friday?

BEHIND THE BUTTON 88W

What a revision costs. A basket changed between two months makes those two months incomparable, so a revision needs both baskets priced in an overlap month. That is real work and it is the reason revisions are rare and overdue.

What refusing to revise costs. An index whose weights are wrong reports the price of a household that does not exist. Coal at a fifth of spending understates a city that now buys electricity and imported flour, and it does so in whichever direction coal happens to move.

TAKES AS READ

the stall prices and the queue are representative of household spending · gross domestic product, and the four things it adds up — taken as read

VERDICT 108W

Both objections are real: a basket has to be fixed for two months to be comparable, and this one has weights from a city that has changed. Pricing both baskets in the same month settles it. The overlap is what links the old series to the new one. It costs one month of double work rather than the whole series. Keeping it preserves comparability with a figure nobody should be comparing. Dropping coal without reweighting leaves the remaining weights summing to four fifths, so every number afterwards is wrong by a quarter. And one stall is one household's basket, which is the error the index exists to avoid.

TAKEAWAY 12W

The weights are the claim an index makes about how people live.

4.2 The account and its mirror

WIRE ROOM • A PERSON • BALLPARK

SCENE 27W

Levon Sarkis, the trade and reserves officer, has the month's wires: exports 64 million, imports 121, net income from abroad minus 8, and remittances in of 19.

GUIDE 58W

Add the four flows with their signs. Two of them are the trade balance and two are the parts people forget, and one of those is large enough here to change the conclusion. Then say what a deficit on this account has to be financed by, because that is what makes it a constraint rather than a statistic.

ASK 6W

Estimate the month's current account balance.

BEHIND THE BUTTON 88W

What remittances do to a small economy. Halvern has people working abroad and sending money home, and 19 million a month is a third of the trade gap. An analysis that looks only at exports and imports understates the country's position by exactly that.

Why the account is a constraint. A deficit means the country is buying more from abroad than it is selling. The difference has to be paid for: by borrowing, by selling assets, or by running down reserves. On the fifteenth it will be reserves.

TAKES AS READ

the four figures are all for the same month • gross domestic product, and the four things it adds up — taken as read

VERDICT 107W

Exports less imports is a gap of 57 million. The two terms people leave out change it substantially. Eight million paid abroad in income makes it worse, and nineteen million of remittances makes it better, for a deficit of 46. That is a third of the gap closed by money Halvern's own people send home, and an analysis that stops at trade misses it entirely. The reason the number matters on Friday is what a deficit has to be financed by. There is no borrowing available, so it is reserves, and reserves divided by 46 million a month is a number of months rather than an opinion.

TAKEAWAY 13W

A current account deficit has to be financed by borrowing or by reserves.

4.3 Two counts of the same money

NOTE ROOM • A ROOM • PROTOCOL

SCENE 24W

The weighed count says 228,000 notes returned. The six regional ledgers, added, say 241,000. Femi Adeyinka, the vault officer, will not sign either figure.

GUIDE 49W

Two counts of the same thing disagree by thirteen thousand notes. Work out what each method would get wrong, and which of the four explanations is consistent with the weighed figure being the lower of the two. One of them is an error and one of them is money.

ASK 11W

Match each explanation to what it predicts about the two counts.

BEHIND THE BUTTON 87W

What the weighing can get wrong. The mass per note comes from a sample, so a mix with more heavy notes than sampled makes the count too low. Thirteen thousand out of 241,000 is about five per cent, which is well within what a bad sample could do.

What the ledgers can get wrong. A note counted at a branch and again at the centre is double-counted, which pushes the ledger total up. Both explanations fit the direction, and only a recount of one region separates them.

TAKES AS READ

both counts cover the same returns • a price index, and the basket it is an index of — taken as read

VERDICT 112W

Four explanations, and each predicts something different that can be checked without arguing. A sampled mass per note set too high divides the same mass into too few notes. So the weighed figure is low, and a recount of one region settles it. A region counting twice inflates the ledger and its own dispatch note will not match its tally. If both methods are right, the thirteen thousand notes left between the branch and the centre, which is a matter for somebody other than the statisticians. And a scale reading low would show as the same proportional gap on every weighing this fortnight, which is a single test on this morning's returns.

TAKEAWAY 14W

When two methods disagree, the question is what each of them would get wrong.

4.4 Forty people, and what they have heard — Review

COUNTER ROOM · A ROOM · CALLBACK · TRIAGE

SCENE 38W

Ngozi Okonjo, the counter supervisor, has a rumour on the plaza that the counter shuts at noon on Friday. It is not true. The queue has doubled again since eight and nobody at the window can say otherwise.

GUIDE 58W

Four things the board could put on the plaza, and each is a statement about the future that the queue will act on this morning. Ask of each whether it gives people something they can check afterwards. Then ask what it costs the board on the fifteenth if the thing they checked turns out to have been wrong.

ASK 10W

What should the board put on the plaza this morning?

BEHIND THE BUTTON 95W

Why a promise is an instrument rather than a courtesy. People queue on what they expect, so the belief moves before any policy does. A statement that can be checked later either earns the board a cheaper way to move expectations next time or spends one it already had.

Why silence is not neutral. The plaza will have an expectation whether or not the board supplies one, and it will be supplied by whoever is talking. The choice is not between saying something and saying nothing; it is between the board's account and the rumour's.

TAKES AS READ

the rumour is false and the counter will open as usual on Friday · the short-run trade-off between inflation and unemployment — taken as read

VERDICT 86W

A denial is a claim about a rumour rather than about the counter, and it leaves people with nothing to hold the board to. A dated schedule does: anybody can check it on Friday, and a promise that survives being checked is what makes the next one worth acting on. Open as long as there is a queue cannot be kept on a day like the fifteenth. A promise broken once costs more than the silence it replaced. Waiting concedes the plaza to whoever is talking.

TAKEAWAY 13W

A statement people can check afterwards moves expectations; one they cannot is noise.

END OF THE DAY 15W

A basket has to be fixed to be comparable and revised to be about anybody.

A column that multiplies

PLAN CARD 99W

Friday, and the board meets at four. Sunita Iyer has argued the same point all week. The reserve requirement has to rise before the new currency goes out, not after. Ruth Haldane has the arithmetic for either timing. She can say what each one does to her loan book. Verhoeven still wants a rate today. Today you work out what a reserve injection can grow into, and what a requirement caps it at. The money supply is not a quantity anybody chose. It is what the banks made. The requirement is the only part of it the board holds.

BRIEFING 12W

Iyer wants the reserve requirement raised before the new notes go out.

WHAT YOU DECIDE 12W

Work out what the banking system can create, and what caps it.

5.1 One over the ratio

LEDGER HALL • A ROOM • BALLPARK

SCENE 21W

The requirement is two per cent. Iyer wants eight. Haldane's bank holds 21 million of reserves against 780 million of deposits.

GUIDE 38W

Work out the multiplier at each requirement and compare them. Then look at Haldane's own numbers. A bank already lent up against a two per cent requirement has to find reserves or unwind loans when the requirement moves.

ASK 7W

Estimate the money multiplier at each requirement.

BEHIND THE BUTTON 94W

Why one over the ratio. Each pound of reserves can support deposits up to the point where the required fraction of them equals that pound, so the maximum is the reciprocal. At two per cent a pound supports fifty; at eight it supports twelve and a half.

Why the timing is Iyer's whole argument. Raising the requirement before the new notes go out constrains lending that has not happened. Raising it afterwards forces banks to call in loans already made, which contracts the money supply exactly when the board is trying to stabilise it.

TAKES AS READ

banks lend up to the limit the requirement allows • fractional reserve banking: a deposit is somebody else's loan — taken as read • the spending multiplier, and why one round is not the total — taken as read

VERDICT 114W

One over 0.08 is 12.5, and one over 0.02 is 50: quadrupling the requirement cuts the multiplier to a quarter of what it was. That is why a change that sounds small — six percentage points — is the largest single lever on the money supply the board has. It also settles Iyer's timing argument. Haldane's bank holds 21 million against 780 million of deposits, which is 2.7 per cent. So at eight per cent she is 41 million short, and has to find reserves or call in loans. Doing that before the new notes are issued constrains lending that has not happened yet; doing it after contracts the money supply during a changeover.

TAKEAWAY 17W

The multiplier is the reciprocal of the reserve ratio, so a small change is a large one.

5.2 Twelve million, or six hundred

RATE ROOM • A PERSON • BALLPARK

SCENE 25W

The treasury wants 12 million of reserves put into the banks to keep wages paid through the changeover. The requirement is still two per cent.

GUIDE 45W

Work out what twelve million of reserves can become in deposits at the current requirement, and then at Iyer's proposed one. The gap between the two answers is the argument the board is about to have, and it is much larger than the injection itself.

ASK 13W

Estimate the deposits 12 million of reserves could create at two per cent.

BEHIND THE BUTTON 85W

Why the injection is not the increase. Reserves lent become deposits, which are lent again less the required fraction, and so on. The total the system can create is the injection times the multiplier, and only the multiplier is under the board's control.

What makes it a ceiling rather than a forecast. Banks lend to the limit only if there are borrowers and confidence. In a changeover fortnight there may be neither, so the figure is the most that can happen rather than what will.

TAKES AS READ

banks lend up to the limit at whatever requirement is in force • the money market: a rate that clears the demand for money — taken as read

VERDICT 98W

Twelve million times a multiplier of fifty is 600 million marks of deposits, which is fifty times the injection and about a quarter of Halvern's entire deposit base. At Iyer's eight per cent the same twelve million supports 150 million, and the difference — 450 million — is the whole of the argument about timing. It is a ceiling rather than a forecast: banks only lend to the limit if there are borrowers, and in this fortnight there may not be. But a board that injects reserves without knowing the ceiling has decided something it has not calculated.

TAKEAWAY 14W

An injection times the multiplier is a ceiling on the money it can become.

5.3 What the queue has already decided

COUNTER ROOM • A ROOM • CHOICE

SCENE 26W

Okonjo's queue is holding at forty. Of the last hundred people, eighty-one took the maximum they were allowed and nineteen took less than they could have.

GUIDE 48W

Look at what people did rather than at what they said. Taking the maximum allowed is a bet on one thing and taking less is a bet on the other. Work out which way the queue has voted and what that means for a rate announced this afternoon.

ASK 6W

What has the queue already decided?

BEHIND THE BUTTON 84W

What taking the maximum means. It is a bet that the old currency is worth less later than now, which is a bet on the rate moving against it. Eighty-one in a hundred is not a rumour; it is a distribution.

Why it constrains the board rather than informing it. A rate set above what people expect will be believed and one set below it will be tested, and the testing is done with reserves. Expectations are cheap to use and expensive to contradict.

TAKES AS READ

the hundred people are the most recent at the counter • the short-run trade-off between inflation and unemployment — taken as read

VERDICT 121W

Eighty-one people in a hundred took the maximum they were allowed. That is a bet that waiting costs them, and that the old currency buys less later than now. That is a statement about the expected rate rather than about the counter, and eighty-one in a hundred makes it a distribution rather than a rumour. It matters this afternoon because a rate set above what people expect is believed and one set below it is tested with reserves. Closing hours would change when people come rather than how much they take. A shortage of new notes would show as people taking less. And a board expected to set no rate would produce hoarding of goods rather than a queue for currency.

TAKEAWAY 13W

What people do at a counter is a distribution rather than a rumour.

5.4 The same list, twice — Review

PRICE ROOM · A ROOM · CALLBACK · PROTOCOL

SCENE 36W

Emil Radic, the chief statistician, keeps a second index for the port: a fixed list of wholesale goods, priced monthly. Four things happened to it this month, and the board wants each one accounted for separately.

GUIDE 59W

Four events and four effects, on an index that prices something else entirely. Pair them by asking one question of each event. Does it move a price on the list, a price off the list, or the year the ratio is measured from? Where a listed price moves, its weight decides how much of that movement reaches the number.

ASK 11W

Match each event to what it does to the wholesale index.

BEHIND THE BUTTON 110W

Why weight is the whole of it. An index is one number standing for a list, and each item enters in proportion to what households or firms spend on it. A fifth of the weight carries a fifth of its own price change into the index, which is why arguing about the weights is arguing about the figure.

What re-basing is and is not. Setting the base year to this month divides every future cost by this month's cost instead of the old one. So the series reads 100 again without a single price having moved. It changes what the number is measured from and never what it is measuring.

TAKES AS READ

the wholesale list is held fixed between the two months · each item enters the index in proportion to its weight

VERDICT 87W

Only a price on the list can move the number, and it moves it in proportion to its weight. So coal at a fifth of the basket carries a fifth of its own doubling into the index. A price the list does not name is invisible however far it moves. Re-basing changes nothing except where the ratio is measured from, so the index reads a hundred again with every price where it was. And substitution the fixed list cannot see is the cost of holding it fixed.

TAKEAWAY 17W

An index moves only for prices on its own list, and only in proportion to their weight.

END OF THE DAY 15W

A pound of reserves supports several pounds of deposits, and the requirement sets how many.

What the wages become

PLAN CARD 97W

Saturday. Josiah Mbeya has brought the treasury's plan. Pay two weeks of public wages early, so that nobody goes unpaid across the changeover. He is straightforward about it. It is the right thing to do, and it makes the inflation figure worse. Radic wants the size of that effect before the board votes. Today you work out what the payment becomes once it has been spent and respent. You also say which curve it moves. Money paid out once is spent several times over. The several times is the part that shows up in a price index.

BRIEFING 12W

The treasury wants to pay wages early, and every option moves demand.

WHAT YOU DECIDE 12W

Work the spending multiplier and say which curve the treasury's plan moves.

6.1 Out of every extra mark

RATE ROOM · A ROOM · BALLPARK

SCENE 23W

Prentice's household survey: when disposable income rose 40 marks a month last year, measured consumption rose 30. Mbeya's plan pays out 84 million.

GUIDE 48W

The fraction of an extra mark that gets spent rather than saved is what makes a multiplier large or small. Work it out from the survey. Note that the rest of it is not lost but saved, which matters for the loanable funds argument later in the fortnight.

ASK 6W

Estimate the marginal propensity to consume.

BEHIND THE BUTTON 87W

Why it is measured rather than assumed. A propensity to consume is a behaviour and it changes with who is being paid and what they expect. During a changeover people spend more of an extra mark than usual, because holding the old currency is a losing bet.

Why the survey is the weak point. Last year's behaviour at a stable price level is not this fortnight's. The figure is the best available and the board should be told it is a lower bound rather than an estimate.

TAKES AS READ

the survey's households are representative of those being paid · gross domestic product, and the four things it adds up — taken as read

VERDICT 108W

Thirty over forty is 0.75, so three quarters of every extra mark is spent. The remaining quarter is saved rather than lost, which is what the loanable funds market later in the fortnight is about. The number matters because it sets the multiplier. At 0.75 the multiplier is four, and at 0.6 it would be two and a half. So a survey figure is doing a great deal of work. It is also the weakest link. Last year's behaviour at stable prices is not this fortnight's, when holding the old currency is a losing bet. So 0.75 is a lower bound, and the board should be told so.

TAKEAWAY 12W

The propensity to consume is a measured behaviour rather than a constant.

6.2 One over one minus it

PRICE ROOM · A PERSON · BALLPARK

SCENE 23W

The propensity to consume is 0.75. The treasury's payment is 84 million marks. Mbeya's paper describes the effect on demand as 84 million.

GUIDE 41W

The payment is spent, becomes somebody's income, and three quarters of it is spent again. Work out the total using the multiplier, then compare it with the paper's figure. The difference is not a correction; it is most of the effect.

ASK 6W

Estimate the total effect on demand.

BEHIND THE BUTTON 84W

Where the series comes from. Eighty-four million, then three quarters of that, then three quarters of the remainder, and so on. The sum of the series is the payment divided by one minus the propensity, which is why the formula is a reciprocal.

Why the total is a ceiling. Each round takes time, and the fortnight is fourteen days. The multiplier is the eventual effect, and how much of it lands before the fifteenth is a separate question the board also has to ask.

TAKES AS READ

the propensity holds across the rounds of spending · aggregate demand, and the four components that move it — taken as read

VERDICT 111W

At a propensity of 0.75 the multiplier is one over a quarter, which is four, so 84 million becomes 336. The paper's figure is the first round of spending and nothing else, and the rounds after it are three times as large again. That is the difference between a payment and its effect on demand. It is the number that has to be in front of the board before it votes. Multiplying by the propensity instead gives 63 million, which is the second round on its own. The honest caveat is time: the multiplier is the eventual total, and how much of it lands inside fourteen days is a further question.

TAKEAWAY 18W

A multiplier is the sum of a series, so the first round is a fraction of the total.

6.3 Where some of it goes

WIRE ROOM • A ROOM • CHOICE

SCENE 24W

Halvern imports its flour, its fuel and most of its manufactures. The payment is 84 million and the multiplier makes the demand effect 336.

GUIDE 45W

Ask where extra spending in Halvern actually lands. Some of it buys local services and some of it buys imported goods, and the imported part leaves the country rather than circulating in it. Say what that does to the multiplier and to the current account.

ASK 10W

What does the import share do to the treasury's plan?

BEHIND THE BUTTON 83W

Why imports leak out of the multiplier. A mark spent on imported flour becomes income abroad rather than in Halvern, so it does not come back for another round. The larger the import share, the smaller the multiplier actually is.

What it does to the account. The same spending widens the current account deficit, which is financed by reserves — the same reserves that would be defending a rate. So the fiscal plan and the exchange rate policy are competing for one pot.

TAKES AS READ

the import share of extra spending is similar to the average • gross domestic product, and the four things it adds up — taken as read

VERDICT 104W

A mark spent on imported flour becomes income abroad. So it does not return for another round of domestic spending. The multiplier is smaller than one over one minus the propensity to consume suggests. The same mark also appears as an import in the current account, which widens the deficit. Both effects run the same way and both matter on the fifteenth, because the deficit is financed by the reserves that would otherwise be defending a rate. The plan therefore does less for demand and more to the account than the paper says, which is a stronger objection than either half on its own.

TAKEAWAY 13W

Spending on imports leaves the domestic circuit and lands in the current account.

6.4 Somebody else's loan — Review

LEDGER HALL · A ROOM · CALLBACK · SEQUENCE

SCENE 35W

Ruth Haldane, who manages the largest of the eleven regional banks, has a single ten-thousand-mark deposit walked in this morning. Iyer wants the four ledger entries it produces set out in the order they happen.

GUIDE 51W

Four entries from one deposit. Order them in time, and settle each by asking what has to have been written before it can be written at all. Three of the four are on Haldane's own ledger and one is not, which is a hint about where the last of them belongs.

ASK 11W

Put the four ledger entries in the order they are written.

BEHIND THE BUTTON 108W

Why only one of the four is a decision. The deposit arriving and the payment leaving are things that happen to the bank. The split between what is held and what is lent is the one entry somebody chooses. That is exactly why a reserve requirement is a policy tool and the other three entries are not.

What the third entry actually creates. A loan is not handed over in cash; it is credited to an account, and an account balance is a deposit. So the borrower has money that did not exist that morning, and the bank's books balance because the loan is an asset against it.

TAKES AS READ

the bank holds a fraction of each deposit and lends the rest

VERDICT 85W

Nothing can be set aside out of money the bank does not yet hold, so the deposit is written first. The split into reserves and lendable funds comes next, and it is the only entry that is a decision rather than a consequence. The loan follows, and the thing worth noticing is that it is credited as a deposit. The builder now has an account balance that did not exist this morning. When he spends it, the same balance turns up at a second bank.

TAKEAWAY 15W

A loan is written as a deposit, which is why lending and deposits grow together.

END OF THE DAY 17W

A pound spent becomes income and is partly spent again, so one round is not the total.

What the reserves can hold

PLAN CARD 115W

Sunday, and the board sits again at six. Verhoeven wants a fixed rate announced with the new notes. Levon Sarkis has the reserve book. He will not give her a number without a period attached to it. Farr points out the risk. A rate announced and then abandoned costs more than never fixing one at all. Today you choose the rate the reserves can actually hold. That means turning the reserve stock into a number of days at each candidate rate. It also means settling what the announcement has to carry beyond the figure. A stock divided by a daily outflow is a deadline. A peg is a promise with that deadline already inside it.

BRIEFING 11W

Verhoeven wants a fixed rate and Sarkis has the reserve position.

WHAT YOU DECIDE 11W

Price the currency and say how long a defended rate lasts.

7.1 A stock over an outflow

WIRE ROOM • A ROOM • BALLPARK

SCENE 23W

Sarkis has 187 million marks of foreign reserves. At the rate Verhoeven wants, the net outflow is running at 5.2 million a day.

GUIDE 51W

Divide the stock by the daily outflow to get a number of days, then count how many days the rate has to hold for. A peg that lasts less than the period it is promised for is worse than no peg, because the abandonment is a policy event of its own.

ASK 10W

Estimate how many days the reserves would hold the rate.

BEHIND THE BUTTON 83W

Why defending costs reserves daily. Holding a rate the market disagrees with means buying your own currency with foreign reserves every day, at whatever volume is offered. The outflow is not a fee; it is the price of the promise.

What abandoning one costs. Everybody who held the currency on the strength of the promise loses, and the next promise is not believed. Farr's point is that credibility is spent once, so a rate should only be announced if it can be held.

TAKES AS READ

the daily outflow holds at the rate being proposed • an exchange rate as a price, and what moves it — taken as read • monetary policy: the three tools and what each one moves — taken as read

VERDICT 114W

One hundred and eighty-seven million over 5.2 million a day is thirty-six days, and the rate has to hold for ninety-six. So the peg Verhoeven wants is not a policy. It is a policy with a date on it, and the date is inside the period it was announced for. That is the argument against announcing it. The abandonment is itself an event.

Everybody who took the currency on the strength of the promise loses, and the next promise is discounted. Using the monthly deficit as the outflow gives four years, and it is the wrong quantity. The deficit is financed partly by other flows. The defence outflow is what the peg specifically costs.

TAKEAWAY 16W

A reserve stock divided by a daily outflow is how long a promise can be kept.

7.2 Not the rate anybody wants

RATE ROOM • A PERSON • TRIAGE

SCENE 29W

Three candidate rates: 3.60 old marks to the new, 4.15, and 4.80. The outflow at each is 5.2, 1.4 and nil million a day. The reserves are 187 million.

GUIDE 50W

Work out how long the reserves hold at each candidate and compare it with the ninety-six days the rate has to survive. Then choose, and say what the chosen rate costs the people it lands on — because two of the three are affordable and they are not equally fair.

ASK 6W

Which rate should the board announce?

BEHIND THE BUTTON 85W

Why a weaker rate costs less to defend. Setting the new currency cheaper against foreign money means the market is not pushing against the peg, so nothing has to be bought to hold it. The cost moves from the reserves to the households buying imported flour.

Who each rate falls on. At 3.60 savers keep more of what they had and importers pay more of the difference through a devaluation later. At 4.80 importers pay immediately and anybody holding old marks takes the loss now.

TAKES AS READ

the outflow figures hold at each candidate rate • the balance of payments: the current account and its mirror — taken as read • demand, supply and the price that clears a market — taken as read

VERDICT 114W

At 3.60 the reserves last thirty-six days against ninety-six needed, so that rate is an announcement with its own failure date in it. At 4.15 the outflow is 1.4 million a day and 187 million lasts about 134 days, which covers the period with a margin. At 4.80 nothing has to be defended, and the whole adjustment lands at once on everybody buying imported flour. That is affordable for the board and expensive for the city. Floating is defensible and gives up the one thing the fortnight is for: something for wages and contracts to be written against. The choice is the affordable rate that costs the least to the people it lands on.

TAKEAWAY 15W

A rate that cannot be defended is a promise with its own expiry inside it.

7.3 A number with a reason

COUNTER ROOM • A ROOM • TRIAGE

SCENE 29W

Farr has a draft announcement with the rate in it and nothing else. Okonjo has forty people who will read it in the morning and decide what to do.

GUIDE 38W

A rate is a number and an announcement is an instrument. Decide what the statement has to carry beyond the figure. The people reading it are choosing whether to hold the new currency or spend it at once.

ASK 10W

What has to be in the announcement besides the rate?

BEHIND THE BUTTON 88W

Why the reason travels further than the number. A rate announced with the reserve position behind it can be checked, and a rate announced alone can only be believed or not. What people are deciding is whether the rate will hold, which is a question about the reserves rather than about the figure.

What must not be in it. Anything the board might have to correct. A single revision costs more credibility than any sentence buys, which is Farr's one rule and the reason the statement is short.

TAKES AS READ

the statement is published before the counter opens • the short-run trade-off between inflation and unemployment — taken as read

VERDICT 106W

What the queue is deciding is whether the rate will hold, and that is a question about reserves rather than about the number. Publishing the position lets anybody check the promise instead of taking it, which is what makes it hold. A price forecast is exactly the figure the board might have to correct, and one revision costs more than the forecast buys. An assurance that the rate will never change is a promise the board cannot keep and will be tested on. Naming the votes is a transparency measure about the board rather than evidence about the currency, and it moves nobody in the queue.

TAKEAWAY 16W

A rate announced with its reason can be checked; one announced alone can only be believed.

END OF THE DAY 12W

Defending a rate spends reserves daily, so the stock is a clock.

Who the difference falls on

BEFORE THE DAY — HUNT

Six ledgers, and one of them does not reconcile

Femi Adeyinka, the vault officer, has six regional ledgers out across these four floors and a weighed total that does not match their sum. Find all six and read them. The gap between the weighed returns and the ledgers is either an accounting error or several hundred thousand marks that never came back.

PLAN CARD 100W

Monday. Ruth Haldane has brought her loan book upstairs. Every loan in it was written at a nominal rate. All of those rates were agreed before prices started moving. Eleven per cent a month has turned careful lending into something else. Iyer wants to know which of the eleven banks is solvent on Friday. Mbeya wants to know whether the treasury's own borrowing has got cheaper. Today you work out the real rate and say who it moves money between. Inflation nobody expected does not destroy money. It moves it, and it moves it one way rather than the other.

BRIEFING 12W

Eleven banks have loan books written at rates set before the changeover.

WHAT YOU DECIDE 13W

Work the real interest rate and say who gains and loses from it.

8.1 Nine a year, against eleven a month

LEDGER HALL • A ROOM • BALLPARK

SCENE 28W

Haldane's loan book is written at 9 per cent a year. Prices are rising at 10.9 per cent a month, which is about 250 per cent a year.

GUIDE 42W

Subtract inflation from the nominal rate over the same period. Mind that the two figures are quoted on different clocks: one is annual and one is monthly. Then say who is better off for the difference, because the money does not vanish.

ASK 9W

Estimate the real interest rate on the loan book.

BEHIND THE BUTTON 98W

Why the period has to match. Nine per cent a year against 10.9 per cent a month is not a subtraction until both are on the same basis. Compounded, 10.9 per cent a month is about 250 per cent a year, and that is the figure the nominal rate has to be set against.

Who gains. The borrower repays in money worth far less than the money borrowed, so the lender's real return is deeply negative and the borrower's real cost is too.

Unexpected inflation transfers wealth from lenders to borrowers, and Haldane's depositors are the lenders here.

TAKES AS READ

the monthly inflation rate persists for the year • the loanable funds market, and where saving meets investment — taken as read • nominal against real: taking the price change out of a figure — taken as read

VERDICT 101W

Nine per cent a year against 250 per cent inflation is a real rate of about minus 241 per cent. Every loan in the book is being repaid in money worth a fraction of what was borrowed. So the borrowers are hugely better off. Haldane's depositors, who are the ultimate lenders, carry the whole loss. That is what unexpected inflation does: it does not destroy the money, it moves it, and always in that direction. Subtracting the monthly figure from the annual one gives minus 1.9 per cent, which mixes two clocks and makes a catastrophe look like a rounding error.

TAKEAWAY 11W

An unexpected rise in prices moves wealth from lenders to borrowers.

8.2 The same debt, in different money

RATE ROOM • A PERSON • CHOICE

SCENE 25W

The treasury owes 340 million marks at a fixed 7 per cent, borrowed before the changeover. Mbeya wants to know whether that is now cheap.

GUIDE 37W

Ask what the debt is denominated in and what has happened to that money. Then say what follows for the treasury and for whoever holds the debt, and whether the answer is a reason to borrow more.

ASK 10W

What has happened to the treasury's borrowing, and what follows?

BEHIND THE BUTTON 84W

Why the debt has become cheap. It is fixed in old marks and old marks are worth far less than when it was borrowed, so the real burden has collapsed. The treasury gains exactly what the holders of the debt lose.

Why that is not a strategy. It works once and only because it was unexpected. The next lender prices in what just happened, so the government borrows at a nominal rate that contains the inflation. The trick is not available a second time.

TAKES AS READ

the debt is denominated in old marks at a fixed rate • the spending multiplier, and why one round is not the total — taken as read

VERDICT 97W

The debt is fixed in old marks, and old marks have lost most of their value. So the real burden has collapsed. The treasury gains precisely what the holders of the debt lose. What does not follow is borrowing more. The gain came from the inflation being unexpected. The next lender prices in what has just happened, so new borrowing carries a nominal rate with the inflation inside it. A fixed nominal rate is exactly what makes the real cost fall, so "unchanged" has it backwards, and rising prices make a fixed-rate debt cheaper rather than dearer.

TAKEAWAY 13W

Inflating away a debt works once, because the next lender prices it in.

8.3 One index, many households

PRICE ROOM · A ROOM · TRIAGE

SCENE 38W

Iyer wants to deflate eleven loan books by the index. Radic points out that the index is one basket, and a bank lending to farmers is not lending to the same prices as a bank lending to importers.

GUIDE 35W

The index is an average over one basket. Ask what that makes it good for and what it cannot settle, then choose the honest way to use it on eleven books that face different prices.

ASK 11W

How should the index be used on the eleven loan books?

BEHIND THE BUTTON 83W

What a single index is good for. Comparing one period with another on a consistent basis. That is exactly what deflating a loan book needs. It is why the index is the right tool even though it is not the right basket for anybody in particular.

What it cannot settle. Whether a particular bank's borrowers face those prices. A farmer selling grain into a rising market and an importer buying flour in a falling currency are on opposite sides of the same index.

TAKES AS READ

the index is the only price series available · a price index, and the basket it is an index of — taken as read

VERDICT 111W

The index is the right tool for the job it is being asked to do, which is comparing one period with another on a consistent basis. It is the wrong basket for any particular borrower. So the honest use is to apply it consistently and say what it is: one basket, an average, not a statement about farmers or importers specifically. Deflating only some books makes them incomparable with the others. Eleven separate indices is a year of work Radic does not have and would produce eleven series nobody can add up. And using nominal figures avoids the caveat by discarding the correction entirely, which is worse than an imperfect one.

TAKEAWAY 12W

One index is right for comparing periods and silent about which household.

END OF THE DAY 15W

The real rate is the nominal rate less inflation, and the difference has an owner.

How much money there is

PLAN CARD 105W

Tuesday. The board publishes a money supply figure on Thursday. There are three counts to work from. Tulloch has weighed returns from the note room. Adeyinka has the vault ledger. Iyer has eleven bank balance sheets. The three of them overlap, and nobody has added them up without counting something twice. Iyer still wants the requirement raised before the new notes go out. This is the day she gets a number for it. Today you count the money and decide what caps it. A money supply figure that counts a vault holding twice is a figure the board will be asked about for a year.

BRIEFING 15W

The weighed returns, the ledgers and the deposits have to add up to one figure.

WHAT YOU DECIDE 8W

Reconcile the money supply and decide the requirement.

9.1 In hands, and in ledgers

NOTE ROOM • A ROOM • BALANCE

SCENE 23W

Issued old currency: 1.42 billion marks. Returned and burnt: 214 million. Held in the vault, never issued: 96 million. Bank deposits: 2.4 billion.

GUIDE 45W

Money is what people can spend: cash in hands plus what banks owe on demand. Read the panel and report the figure. Two of these numbers are not money in the public's hands at all, and one of them is easy to add by mistake.

ASK 9W

Read what you need and report the money supply.

BEHIND THE BUTTON 86W

Why the vault holding is not money. Currency printed and never issued is paper. It becomes money when it leaves the vault, and counting it beforehand overstates the supply by whatever is in stock.

Why the burnt notes come off. They were money and are not any more, so the cash in hands is what was issued less what has come back and been destroyed. Notes returned and not yet burnt are the awkward case and are counted as withdrawn once they are in the office.

TAKES AS READ

deposits and currency figures are for the same date • demand, supply and the price that clears a market — taken as read

VERDICT 110W

Money is what can be spent: cash in the public's hands plus deposits repayable on demand. That is 1,420 million issued, less the 214 million burnt, plus 2,400 million of deposits. The term that does not announce itself is interbank. Of the deposit total, 82 million is banks holding accounts with each other. That appears on two balance sheets and is one deposit, so it comes off. Two more figures are traps in opposite directions. The vault stock has been printed and never issued, so it is paper and adding it overstates the supply. The notes in the note room are already counted as withdrawn, so subtracting them again double-counts.

TAKEAWAY 12W

Printed and unissued currency is paper; issued and unreturned currency is money.

9.2 Two per cent, or eight

LEDGER HALL • A PERSON • TRIAGE

SCENE 26W

The requirement is two per cent. Iyer wants eight before Thursday. Haldane's bank holds 2.7 per cent and three of the eleven hold less than four.

GUIDE 45W

Work out what the change asks of the banks that are furthest from it, then decide the timing. Raising the requirement caps what the system can create and forces anybody below the new figure to find reserves — and the two things happen at once.

ASK 8W

What should the board do about the requirement?

BEHIND THE BUTTON 95W

What eight per cent asks of Haldane. She holds 21 million against 780 million of deposits, and eight per cent of 780 is 62.4 million. That is 41 million she has to find, by selling assets or by calling in loans, and there is no market for either this fortnight.

Why in steps rather than at once. A requirement raised in stages gives banks time to accumulate reserves out of new deposits rather than by unwinding loans. It also caps the multiplier immediately, which is the part that matters before the new notes go out.

TAKES AS READ

the eleven banks cannot raise capital externally this fortnight • fractional reserve banking: a deposit is somebody else's loan — taken as read • the spending multiplier, and why one round is not the total — taken as read

VERDICT 114W

The requirement does two things at once, and they have different clocks. Capping what the system can create matters immediately, before 12 million of reserves becomes 600 million of deposits. Making eleven banks hold more reserves takes time, because three of them would have to find tens of millions by selling assets nobody is buying. Steps do the first straight away and give the second time to happen out of new deposits. Going to eight at once risks three bank failures during a changeover. Leaving it until afterwards forces the contraction at the worst possible moment. And a different requirement for weak banks rewards being weak and tells the market which three they are.

TAKEAWAY 10W

A requirement caps creation immediately and forces adjustment over time.

9.3 Money, output and prices

WIRE ROOM • A ROOM • CHOICE

SCENE 27W

The money supply is 3,606 million and would be 4,206 if the treasury's reserves became deposits at the current requirement. Real output is falling four per cent.

GUIDE 43W

Put the money supply, its turnover, the price level and real output in one relationship and see what has to give. If money grows and output does not, and turnover holds steady, the adjustment happens in prices. Work out the size of it.

ASK 15W

What does a 17 per cent rise in the money supply do, with output falling?

BEHIND THE BUTTON 89W

What velocity is and why it is assumed. How many times a unit of money is spent in a period. It is not constant, and in a changeover it rises because nobody wants to hold the old currency — which makes the price effect worse rather than better.

What the relationship does and does not prove. It is an identity: the four terms are defined so that it holds. What makes it useful is the assumption about velocity, and that assumption is where every argument about it actually is.

TAKES AS READ

velocity holds steady over the fortnight • fractional reserve banking: a deposit is somebody else's loan — taken as read • the spending multiplier, and why one round is not the total — taken as read

VERDICT 104W

The relationship holds money times its turnover equal to prices times real output. If money rises 17 per cent and turnover holds, the right-hand side must rise 17 per cent too. And real output is falling four per cent. So prices have to rise by more than 17 to make up the difference. That is the arithmetic, and the real situation is worse. Velocity rises in a changeover because nobody wants to hold the currency, which adds to the left-hand side again. Prices rising by exactly the money growth requires output to be flat. Nothing here makes output recover; money does not create capacity.

TAKEAWAY 14W

If money grows faster than output and turnover holds, the difference lands in prices.

END OF THE DAY 13W

The money supply is cash in hands plus deposits, and neither counts twice.

How far from capacity

PLAN CARD 91W

Wednesday. The board has to say in writing what Halvern's problem is. Is it demand, or is it supply? The two call for opposite policies. Prentice has the labour figures. Radic has the output series. Verhoeven has to sign whichever sentence the numbers support. Mbeya's wage plan is a demand answer. Iyer's requirement is a supply constraint. Today you measure the gap between actual output and potential output, and say what it implies. A policy aimed at the wrong side of the diagram makes the problem it was aimed at worse.

BRIEFING 10W

Two people are reading the same figures as opposite problems.

WHAT YOU DECIDE 11W

Measure the gap between what Halvern makes and what it could.

10.1 Actual, against potential

PRICE ROOM · A ROOM · BALLPARK

SCENE 30W

Real output is running at 629 million a month. Radic's estimate of potential, at full employment, is 712 million. Unemployment is 8.4 per cent against a natural rate of 5.

GUIDE 42W

Work out how far actual output is from potential as a percentage of potential. Then check it against the labour figures. A gap in output and a gap in unemployment should tell the same story, and here they can be compared directly.

ASK 4W

Estimate the output gap.

BEHIND THE BUTTON 85W

What potential output is. What the economy could produce with everybody employed who wants to be, at the capital and skills it has. It is an estimate rather than a measurement, and Radic's has a range on it he will quote if asked.

Why the two gaps agree. Three and a half points of unemployment above the natural rate is a lot of idle labour, and idle labour is missing output. The two figures being consistent is the strongest evidence either of them is right.

TAKES AS READ

the estimate of potential output is of the same period · frictional, structural and cyclical unemployment — taken as read · nominal against real: taking the price change out of a figure — taken as read

VERDICT 92W

Eighty-three million short of 712 is 11.7 per cent below capacity. The figure matters because it is consistent with the labour data. Unemployment is 3.4 points above its natural rate, and idle labour on that scale is missing output on this scale. Two independent estimates agreeing is the best evidence either is right. Dividing by actual rather than potential gives 13.2 per cent. That measures the shortfall against the smaller number and overstates it. The gap is conventionally a fraction of what the economy could do, not of what it is doing.

TAKEAWAY 16W

A negative output gap and unemployment above its natural rate are two views of one thing.

10.2 Demand, or capacity

RATE ROOM • A PERSON • CHOICE

SCENE 29W

Output is 11.7 per cent below capacity and prices are rising 10.9 per cent a month. Mbeya reads it as deficient demand; Iyer reads it as a supply shock.

GUIDE 42W

Both readings explain part of the figures and only one explains both. Work out what each would predict about prices and output together, and say which combination Halvern actually has. Then say what that means for the two policies on the table.

ASK 5W

Which reading fits both figures?

BEHIND THE BUTTON 73W

What deficient demand alone would give. Output below capacity and prices falling or flat, because less spending pulls both down. Halvern has the output half and not the price half, so demand is not the whole story.

What a supply shock gives. Output below capacity and prices rising, because the same production costs more. That fits both figures, and it is why a demand stimulus alone would raise prices further without restoring capacity.

TAKES AS READ

the output and price figures are for the same month • short-run aggregate supply, and what a cost shock does to it — taken as read

VERDICT 105W

Deficient demand alone would put output below capacity and prices flat or falling. Halvern has output eleven per cent down and prices up eleven per cent a month. That is the supply pattern. The same production costs more, so less of it happens at a higher price. Demand is weak too, which is what the output gap says. But the price behaviour is coming from costs, so a stimulus alone would raise prices further without restoring capacity. Excess demand would have output above capacity, not below. And the figures are not inconsistent; they are the signature of the case both readings were trying to simplify.

TAKEAWAY 13W

Two figures moving the wrong way together point at a third thing moving.

10.3 Three kinds of not working

COUNTER ROOM · A ROOM · PROTOCOL

SCENE 22W

Of 41,000 unemployed: about 9,000 between jobs, about 14,000 whose trades have gone, and the rest laid off since the changeover began.

GUIDE 44W

Sort the three groups by what would put each of them back to work. One responds to demand, one to time, and one to training or to moving. Only one of the three is what a monetary or fiscal decision this fortnight can touch.

ASK 11W

Match each group to what would put them back to work.

BEHIND THE BUTTON 80W

Why the natural rate is not zero. People between jobs and people whose trades have gone exist in any healthy economy. That is why full employment is defined at about five per cent here rather than at nought.

What the board can actually move. The 18,000 laid off since the changeover are cyclical: demand fell and they went with it, and demand returning brings them back. The other two groups need time or retraining, and no rate decision reaches them.

TAKES AS READ

the three groups are counted from the same survey · unemployment: who is counted, and who is not in the labour force — taken as read

VERDICT 100W

Three kinds of unemployment and three different answers. Nine thousand between jobs need time and information; that is frictional and it exists in any healthy economy. Fourteen thousand whose trades have closed need retraining or somewhere else to live, and no rate decision reaches them within a fortnight. Eighteen thousand laid off since the changeover are cyclical: demand fell and they went with it. They are the group a policy this month can actually move. The five per cent is not a fourth group: it is where the first two sit, which is why full employment is defined above zero.

TAKEAWAY 8W

Only cyclical unemployment answers to a demand policy.

END OF THE DAY 13W

Where output sits relative to capacity is what says which problem you have.

Three tools and a lag

PLAN CARD 105W

Thursday. The board has three instruments and eleven days. Farr wants an announcement. Iyer wants the requirement raised. Mbeya wants the wages paid. Verhoeven has to put those three in an order. Sarkis points out the awkward part. Whatever is chosen lands after the fifteenth rather than before it. Today you match each tool to the thing it actually moves. That means working out how long the requirement takes to bite. It also means saying what the short-run trade-off looks like once a cost shock has moved it. A correct policy that arrives after the event lands on the recovery instead of on the problem.

BRIEFING 12W

Everything the board can do arrives after the day it was decided.

WHAT YOU DECIDE 8W

Choose the instrument, and know when it lands.

11.1 What each one actually does

RATE ROOM · A ROOM · PROTOCOL

SCENE 26W

Three instruments on the board's list: buy or sell government paper, change the reserve requirement, change the rate at which banks can borrow from the board.

GUIDE 38W

Each tool acts on a different quantity and at a different speed. Match them, and note which of the three changes the money supply directly and which change what banks choose to do with the money they have.

ASK 7W

Match each tool to what it moves.

BEHIND THE BUTTON 84W

Why open market operations are the usual tool. Buying paper puts reserves into the system at once and in whatever quantity is wanted, which makes it precise and reversible. It is the instrument used weekly rather than annually.

Why the requirement is used rarely. It is the bluntest of the three: it changes the multiplier for every bank at once and forces adjustment on the ones furthest from it. That is exactly why it is right this fortnight and wrong in a normal month.

TAKES AS READ

the board has functioning markets in government paper · the money market: a rate that clears the demand for money — taken as read

VERDICT 116W

The three conventional tools act on different quantities. Buying paper hands reserves straight to a bank, which changes the supply directly and can be done in any amount and undone next week. The requirement changes what every bank may lend against what it holds. So it moves the multiplier rather than the reserves. It does that bluntly and for everybody at once, which is why it is rarely used and right this fortnight. The borrowing rate changes what reserves cost and therefore what banks choose to hold. And the fourth item is the one nobody lists as a tool, and it acts fastest. An announcement moves expectations before any of the other three have taken effect.

TAKEAWAY 16W

One tool changes reserves, one changes the multiplier and one changes the price of borrowing them.

11.2 Decided Thursday, felt when

LEDGER HALL • A PERSON • CHOICE

SCENE 26W

Iyer wants the requirement raised on Thursday. Haldane asks when it would actually change anything, and nobody in the room has the reporting calendar to hand.

GUIDE 47W

Separate the three lags in any policy: how long to recognise the problem, how long to decide, and how long until the decision has an effect. Work out which of the three is doing the damage here, then say what follows for the timing of the vote.

ASK 8W

Which lag decides when the requirement is felt?

BEHIND THE BUTTON 92W

The three lags. Recognition — the data arrive late and are revised. Decision — the board has to meet and agree. Impact — the instrument takes time to work through. The last is usually the longest and the least discussed.

Why this one is unusually short at one end. The board is meeting daily and the data are the board's own, so recognition and decision are almost nil. The impact lag is a month to the reporting date and nine months to the loan book, which is where the whole delay sits.

TAKES AS READ

the reporting date is the first of next month • monetary policy: the three tools and what each one moves — taken as read • the business cycle, and where an output gap sits in it — taken as read

VERDICT 105W

Recognition and decision are almost nothing here — the board is meeting daily and the data are its own — so the delay is all impact. A requirement raised on Thursday applies from the next reporting date, which is the first of next month, and the loan book it eventually constrains runs nine months. So the cap on new lending is a month away and the adjustment of existing lending is most of a year. That is why the vote has to happen before the new notes go out rather than after. The effect the board wants is a month behind whatever it decides today.

TAKEAWAY 14W

The impact lag is usually the longest and the least discussed of the three.

11.3 Prices against jobs

WIRE ROOM • A ROOM • CHOICE

SCENE 24W

Mbeya's plan would lower unemployment and raise prices. Iyer's requirement would lower prices and raise unemployment. Verhoeven has to pick one before the fifteenth.

GUIDE 50W

In the short run a demand policy moves prices and unemployment in opposite directions, which is what makes this a choice rather than an optimisation. Say what the trade-off actually is here — and what makes Halvern's case unusual, because a cost shock does not sit on the same curve.

ASK 7W

What does the board face this fortnight?

BEHIND THE BUTTON 91W

Why the trade-off exists in the short run. More spending buys more output and more employment, and pulls prices up doing it. Less spending does the reverse. Along that curve a board chooses a point rather than escaping it.

Why a cost shock is worse than a choice. A supply shock moves the whole trade-off, so both numbers get worse at once and no point on the old curve is available. That is Halvern's case, and it is why neither policy on the table is a solution rather than a preference.

TAKES AS READ

both policies act within the fortnight • demand-pull and cost-push inflation, and telling them apart — taken as read • frictional, structural and cyclical unemployment — taken as read

VERDICT 105W

In an ordinary month the board would be choosing a point: more demand for lower unemployment and higher prices, or less for the reverse. A cost shock does something different. It moves the whole relationship, so every available combination is worse than it was, and no point on the old curve exists any more. That is why Mbeya's plan and Iyer's requirement are both defensible, and neither is a solution. One buys jobs with prices; the other buys prices with jobs. The shock has left both worse than either would have been. It matters now rather than later because the impact lag is a month.

TAKEAWAY 12W

Some events give you a choice and some take the choices away.

END OF THE DAY 14W

A policy has a lag, and the lag decides what it actually lands on.

A number, and what it is written against

PLAN CARD 112W

Friday, and the rate is announced at four whatever happens. This is the day Verhoeven is right about. Another week without a number means wages, rents and contracts written against a guess. That is its own kind of inflation. Sarkis has the reserve arithmetic that rules out the rate savers want. Farr has one page to fill. Today you fix the rate. You then convert a wage into the new currency for the notice at the counter. You state what the rate is held with, and say plainly who pays for it. A rate is a price. Every price has somebody standing on each side of it who did not choose it.

BRIEFING 12W

Verhoeven is right today, and the number has to be defensible anyway.

WHAT YOU DECIDE 9W

Fix the rate and say what it costs whom.

12.1 Four fifteen to the old

RATE ROOM • A ROOM • BALLPARK

SCENE 28W

The rate is 4.15 old marks to one new. The tram drivers earn 2,180 old marks a month. Okonjo needs the figure for the counter notice by four.

GUIDE 51W

Divide or multiply, and be sure which. The rate is quoted as old marks per new one, so a wage in old marks becomes a smaller number of new ones. Work it out, then say what the direction of the quote does to anybody who gets it the wrong way round.

ASK 7W

Estimate the drivers' wage in new marks.

BEHIND THE BUTTON 98W

Why the direction matters more than the arithmetic. A rate quoted as 4.15 old to one new and a rate quoted as 0.241 new to one old are the same rate. Multiplying when you should divide gives a wage seventeen times too large, and the counter notice is read by nine hundred thousand people.

What the new figure means. A wage of 525 new marks buys what 2,180 old ones bought this morning, so nobody is better or worse off at the moment of conversion. Everything that happens afterwards depends on what prices do in the new currency.

TAKES AS READ

the rate is quoted as old marks per new mark • the balance of payments: the current account and its mirror — taken as read • demand, supply and the price that clears a market — taken as read

VERDICT 101W

The rate is old marks per new one, so old marks are divided by it: 2,180 over 4.15 is 525 new marks. Multiplying gives 9,047, which is the same rate applied in the wrong direction. It is a figure seventeen times too large on a notice nine hundred thousand people will read. The other route to the same answer is the reciprocal: 2,180 times 0.241 is 525. Having both on the notice is how the direction is made unmistakable. At the moment of conversion nobody is better or worse off; everything that happens next depends on prices in the new currency.

TAKEAWAY 14W

A rate is a price with a direction, and the direction inverts every conclusion.

12.2 Savers, and importers

COUNTER ROOM · A PERSON · CHOICE

SCENE 31W

At 4.15 rather than the 3.60 savers wanted, somebody holding 10,000 old marks gets 2,410 new ones instead of 2,778. The reserves hold 4.15 for 134 days and 3.60 for 36.

GUIDE 42W

Work out who is worse off at 4.15 than at 3.60. Then work out who is worse off if 3.60 is set and abandoned in five weeks. Both groups are real people and the second group is larger, which is the argument.

ASK 11W

What is the honest way to put 4.15 to the queue?

BEHIND THE BUTTON 79W

What savers lose at the weaker rate. About thirteen per cent of what they held, immediately and visibly, which is why 3.60 is what the queue wants and what the board cannot hold.

What everybody loses if a peg fails. The rate goes wherever the market puts it, which is past 4.80 on Sarkis's numbers. Everybody who took the new currency because the board promised 3.60 takes that loss instead. The saver loses more, later, along with everybody else.

TAKES AS READ

the reserve figures hold at each rate · an exchange rate as a price, and what moves it — taken as read · monetary policy: the three tools and what each one moves — taken as read

VERDICT 108W

Both halves have to be said. At 4.15 a saver holding ten thousand old marks gets 368 new marks less than at 3.60. That is a real loss, and the reason the queue wanted the other number. And 3.60 lasts thirty-six days against ninety-six needed. After that the rate goes past 4.80 and the same saver loses more. So does everybody who took the new currency on the strength of the promise. Calling 4.15 the fairest rate available is a claim about fairness the board cannot support. Saying it will improve later is a promise nobody can keep. And the market has not chosen it; the reserves have.

TAKEAWAY 14W

Two groups pay for any rate, and one of them pays later and more.

12.3 One basket, two currencies

PRICE ROOM · A ROOM · TRIAGE

SCENE 31W

On the fifteenth the same basket will be priced in new marks. Radic has to decide what happens to the series that has been running in old marks for eleven years.

GUIDE 38W

The basket is not changing and the currency is. Decide what has to happen to the index so that a month before the changeover can be compared with a month after it. Then say what the alternative costs.

ASK 10W

What has to happen to the index on the fifteenth?

BEHIND THE BUTTON 90W

Why the index can survive a currency change. An index is a ratio, and a ratio is unitless. Prices in new marks divided by base-year prices converted at the same rate give the same index. So the series continues if the conversion is applied consistently to both.

What breaks it. Converting this month's prices and not the base year's, which changes the index by the rate rather than by any price movement. That would print a 76 per cent fall in prices on the fifteenth and nothing would have happened.

TAKES AS READ

the conversion rate is applied to the whole series · gross domestic product, and the four things it adds up — taken as read

VERDICT 105W

An index is a ratio and a ratio has no units, so a currency change cancels — provided it is applied to both halves. Convert this month's basket and the base year's at the same rate and the index is exactly what it was, and eleven years of series survives. Converting only this month divides the index by 4.15. It would print a 76 per cent fall in prices on a day when nothing happened. That is the error that would end the series' credibility. Starting again at 100 throws away eleven years for no reason. Publishing both invites everybody to quote whichever suits them.

TAKEAWAY 17W

An index is a ratio, so a currency change cancels if it is applied to both halves.

END OF THE DAY 18W

A rate that can be held is worth more than a rate that is fair and cannot be.

What the wires say

BEFORE THE DAY — CANVASS

Ask everyone on the counter floor the same question

The board is about to set a rate on the assumption that people will hold the new currency rather than spend it at once. Ngozi Okonjo, the counter supervisor, has forty people who have been on the plaza telephones since eight and will not put words in any of their mouths. Ask each of them one question and count the answers — expectations are a policy instrument, and this is the only instrument that measures them.

PLAN CARD 116W

Saturday. The two industries left in Halvern want opposite things. The textile mills export, and they want the new mark cheap. The bakeries import flour, and they want it dear. Sarkis has the wires and the reserve position. Mbeya has a delegation from each side waiting downstairs. Verhoeven wants one paragraph on why the rate cannot be set for either of them. Today you work out what the rate does to each side. You also say what Halvern is actually better at making. Then you read what the deposit and lending figures say about the rate that would clear the market. One rate is one price, and it lands on two industries in exactly opposite directions.

BRIEFING 13W

The mills want a weaker rate and the importers want a stronger one.

WHAT YOU DECIDE 14W

Work out what the rate does to trade, and who is better at what.

13.1 Cheap for them, dear for us

WIRE ROOM • A ROOM • CHOICE

SCENE 26W

At 4.15 the mills' cloth costs a foreign buyer 12 per cent less than at 3.60, and the bakeries' imported flour costs 15 per cent more.

GUIDE 36W

Work through what a weaker currency does on each side of the border. Then say why it is not a policy for either industry, given that both effects are the same fact seen from two sides.

ASK 10W

Why can the rate not be set for either industry?

BEHIND THE BUTTON 95W

Why both effects are one fact. A weaker currency means foreigners get more of it per unit of theirs, so Halvern's goods are cheaper to them and theirs are dearer to Halvern. The two cannot be separated, and a rate chosen for the mills is a rate chosen against the bakeries.

Which effect is larger here. Halvern imports far more than it exports — 121 against 64 — so the flour effect touches more of the economy than the cloth effect. That is a fact about the trade position rather than an argument about fairness.

TAKES AS READ

the price changes are those implied by the rate alone • the balance of payments: the current account and its mirror — taken as read • demand, supply and the price that clears a market — taken as read

VERDICT 114W

A weaker currency means each unit of foreign money buys more marks, so Halvern's cloth is cheaper abroad and foreign flour is dearer here. Those are not two effects to be balanced. They are one fact seen from either side of the border. A rate chosen to help the mills is necessarily chosen against the bakeries. The two industries are not equally sized: imports are 121 million against exports of 64. So the flour effect touches more of the economy. That is a reason to be careful rather than a reason to pick the other side. The board's mandate is not what is in question, and the rate is being fixed rather than floated.

TAKEAWAY 14W

One rate, two industries, and the effects are the same fact from two sides.

13.2 Better at, or better off doing

PRICE ROOM · A PERSON · CHOICE

SCENE 30W

Halvern can produce cloth or flour. A worker-month makes 40 bolts of cloth or 9 tonnes of flour. Across the border the same month makes 60 bolts or 10 tonnes.

GUIDE 38W

Work out what each country gives up to make one unit of each good, rather than how much of it they can make. The comparison that matters is the ratio inside each country, not the totals between them.

ASK 6W

Which good should Halvern specialise in?

BEHIND THE BUTTON 80W

Why absolute figures mislead. The neighbour is better at both — more cloth and more flour per worker-month. That does not mean Halvern should make neither: trade depends on what each gives up, not on who is faster.

What the ratios say. In Halvern a tonne of flour costs 4.4 bolts of cloth. Across the border a tonne costs 6 bolts. So flour is cheaper in Halvern in the only sense that matters, and cloth is cheaper across the border.

TAKES AS READ

a worker-month is comparable in both places · scarcity, and why every choice has a next-best alternative — taken as read

VERDICT 98W

The neighbour is better at both in absolute terms, and that is not the comparison. In Halvern a tonne of flour costs 4.4 bolts of cloth foregone; across the border it costs 6. So flour is cheaper in Halvern in the only sense that matters, which is what has to be given up. Both countries gain if Halvern makes flour and buys cloth. Being worse at everything does not exclude a country from trade, which is the whole point of the idea. Making both is what a country does when it cannot trade rather than when it can.

TAKEAWAY 15W

Comparative advantage is about what you give up, not about what you are better at.

13.3 The rate that clears it

LEDGER HALL · A ROOM · CHOICE

SCENE 25W

Iyer has deposits rising as people put new marks in and loan applications at a third of last year's. The board's borrowing rate is unchanged.

GUIDE 33W

Two quantities are moving in opposite directions in a market where a price is supposed to clear them. Say what the price is, which way it should move, and what is stopping it.

ASK 9W

What do rising deposits and falling loan applications imply?

BEHIND THE BUTTON 82W

What the market is. Saving is the supply of funds and investment is the demand, and the real interest rate is what clears them. More saving and less borrowing means the clearing rate has fallen.

Why the observed rate has not moved. It is fixed by the board, which is a decision rather than a market outcome. And with inflation at eleven per cent a month the real rate is deeply negative anyway. That is why nobody is saving for the return.

TAKES AS READ

the deposit and application figures are for the same period · the money market: a rate that clears the demand for money — taken as read

VERDICT 110W

More funds offered and less borrowing wanted is a supply rising against a demand falling, and in any market that means the clearing price has come down. Here the price is the real interest rate. The board's nominal rate has not moved, so the market is not clearing at the posted figure. Banks refusing to lend would show as applications made and declined rather than as fewer applications. Patience is not what is happening. With inflation at eleven per cent a month, the real return on a deposit is deeply negative. People are holding new marks because the alternative was worse. And the money supply is rising rather than contracting.

TAKEAWAY 15W

Saving is supply and investment is demand, and the real rate is what clears them.

END OF THE DAY 12W

A weaker currency helps exporters and taxes anybody who buys from abroad.

The day before the fifteenth

PLAN CARD 118W

Sunday. The counter closes at six for the last time in old marks. Tulloch has notes still coming in. His furnace takes four hours a load. Sarkis has reserves that have to cover a first week of unknown demand. Iyer has three banks sitting below the new requirement. Verhoeven wants one page for the morning. Today you decide how what is left of the reserves gets spent. You also count what is still out there in old notes. Then you settle what happens to the three banks below the requirement. A plan for the fifteenth that assumes the fifteenth goes well is not a plan. By six this evening there is no time left to make another one.

BRIEFING 9W

Everything that has not been decided gets decided today.

WHAT YOU DECIDE 17W

Spend the last of the reserves, the last of the notes and the last of the time.

14.1 What the first week has to cover

RATE ROOM · A ROOM · ALLOCATE

SCENE 22W

187 million of reserves. Six calls on them in the first week, and the total asked for is more than there is.

GUIDE 47W

Everything on this list is a real claim and the reserves will not cover them all. Two of the six are what the rate itself is held with, so they are not optional. Spend the pool, and know what the plan cannot answer once it is spent.

ASK 18W

187 million of reserves. Build the first week, and know what it stops you being able to do.

BEHIND THE BUTTON 74W

What defending the rate costs in week one. At 4.15 the outflow is 1.4 million a day, so a week is about ten million. And the first week of a changeover is the week most likely to test it.

Why the note float is not negotiable. If the counter cannot pay out new marks on the fifteenth, the rate is irrelevant: the failure everybody remembers is a closed counter with a queue outside it.

TAKES AS READ

the reserve total cannot be added to this week · the spending multiplier, and why one round is not the total — taken as read

VERDICT 113W

The float is not a choice: a counter that cannot pay out new marks on the fifteenth makes every other decision irrelevant. Defending the rate for the first week costs ten million and is what the announcement was made with, so it is the second thing bought. Flour is the one item on the list the public will notice within days, and a buffer is what makes week two possible at all. That leaves the three weak banks or the early wages, and not both — which is the decision. The honest form is to say which one is not being funded rather than to plan for both and discover it on Thursday.

TAKEAWAY 10W

A reserve spent on everything defensible defends nothing in particular.

14.2 What is still out there

NOTE ROOM · A ROOM · BALLPARK

SCENE 26W

Tulloch has weighed 986 kilograms of returns in total, at 0.94 grams a note. The board issued 1.42 billion marks in notes averaging 340 marks each.

GUIDE 42W

Work out how many notes have come back and compare it with how many were issued. What is left outside is the part the board will still be honouring next year, and it is the number the closing statement has to carry.

ASK 7W

Estimate how many notes are still outstanding.

BEHIND THE BUTTON 84W

What is left out is not lost. Notes held abroad, notes in mattresses and notes destroyed by accident all stay on the board's books as a liability until they are presented or the period expires.

Why the count matters on the fifteenth. The board has to say how much old currency is still outstanding, because that is what it may still have to convert. A figure that turns out to be a third wrong is a figure it will be asked about for years.

TAKES AS READ

the average note is 340 marks and 0.94 grams · demand, supply and the price that clears a market — taken as read

VERDICT 109W

Nine hundred and eighty-six kilograms at 0.94 grams a note is about 1,049,000 notes returned, against 4,176,000 issued. So 3.1 million are still out. At 340 marks each that is around a billion marks. That is a liability rather than a saving: every one of them may still be presented, and the board's closing statement has to say so. Dividing the return by the average note value instead of by its mass mixes a weight with a price. The result is not a count of anything. What the weighing cannot tell anybody is the mix, which is why the average note value is sampled and quoted with the figure.

TAKEAWAY 12W

What has not come back is a liability rather than a saving.

14.3 Three of eleven

LEDGER HALL · A PERSON · TRIAGE

SCENE 33W

The requirement steps to four per cent on the first and six on the first of next month. Three banks hold between 2.6 and 3.1 per cent and cannot raise capital this week.

GUIDE 46W

Work out what the first step asks of the three banks furthest from it. Then decide what the board should do. Lending them reserves and letting them fail are both real options with real costs. The step was chosen to be reachable; check whether it is.

ASK 9W

What should the board do about the three banks?

BEHIND THE BUTTON 81W

What four per cent asks. A bank at 2.6 per cent has to find nearly a percentage and a half of its deposits in reserves inside a fortnight. For the largest of the three that is about eleven million marks.

What lending it to them costs. Fifty-eight million of the reserve pool, which is the same pool the rate is defended with. The two uses compete directly, which is why yesterday's allocation and today's question are the same decision seen twice.

TAKES AS READ

the three banks cannot raise capital externally this week · fractional reserve banking: a deposit is somebody else's loan — taken as read · the spending multiplier, and why one round is not the total — taken as read

VERDICT 116W

The three banks hold about a third of Halvern's deposits between them. So letting them fail in the first week of a new currency is not a supervisory outcome. It is a bank run with a changeover attached. Lending them reserves costs 58 million out of the same pool the rate is defended with. That is the real price and has to be said out loud: a narrower defence and a shorter peg. Exempting them tells the market exactly which three are weak and produces the run without the cost. Raising the requirement on the strong eight caps the multiplier while leaving the weakest banks untouched, which is the opposite of what a requirement is for.

TAKEAWAY 12W

A reserve requirement and a defended rate compete for the same reserves.

END OF THE DAY 17W

A finite reserve spent against an unknown first week is a decision about what is not covered.

The fifteenth

PLAN CARD 108W

Monday the fifteenth. The counter opens at nine in new marks. The old ones stop being money at midnight. Nine hundred thousand people find out this morning what their wages are. Verhoeven has one page to publish. Radic has an index that has to survive the changeover. Sarkis has a rate to defend for ninety-six days. Today you close the last three figures and say what the fortnight actually established. Everything in this building is now one of two things. It is either a measurement with a period on it, or it is somebody's recollection. What is published today decides which of the two the next board inherits.

BRIEFING 16W

The counter opens at nine in new marks and everything either measures or it does not.

WHAT YOU DECIDE 11W

Open in the new currency and say what the fortnight established.

15.1 What the file supports

PRICE ROOM • A ROOM • PROTOCOL

SCENE 27W

Four statements are drafted for this morning. Two are measurements with their conditions attached and two are not, and all four will be quoted for a year.

GUIDE 38W

Read each statement against what was actually done. A measurement carries the conditions it was taken under; a projection carries the assumptions it was computed from. Both can be published and only one can be relied on afterwards.

ASK 8W

Match each statement to what stands behind it.

BEHIND THE BUTTON 67W

What the fortnight measured. A reweighted basket priced in an overlap month. A money supply reconciled across three counts. A reserve position with a daily outflow. An output gap consistent with the labour data.

What it did not measure. What prices will do next month. That depends on the money supply being held, and on what people expect. Neither of those is a measurement of anything yet.

TAKES AS READ

the statements are read as published rather than explained • a price index, and the basket it is an index of — taken as read

VERDICT 102W

Three of the four are measurements and each carries its conditions. The price figure sits on a reweighted basket, priced both ways in one overlap month. The money supply is three counts reconciled, with the interbank double-count taken out. The rate is published with the reserves behind it. The fourth is a projection. What it rests on is a decision rather than a measurement: whether the money supply is held, and what people expect. Publish it with the assumptions beside it. Do not publish it in the same voice as the other three. It is the one that will be quoted back.

TAKEAWAY 15W

A projection published as a measurement is the one figure that will be quoted back.

15.2 Nine o'clock

COUNTER ROOM • A PERSON • CHOICE

SCENE 29W

Okonjo has forty-two people outside at ten to nine, a float for the week and a rate on the notice. Eighty-one in a hundred took the maximum last week.

GUIDE 36W

The float has to last a week and the queue would take it today. Decide what the counter does at nine, given that a limit protects the float and also tells everybody the float is limited.

ASK 7W

What should the counter do at nine?

BEHIND THE BUTTON 84W

What a daily limit does. It makes the float last, and it signals that the board expects a run — which is itself a reason to join the queue. The signal and the protection come together and cannot be separated.

Why publishing the float changes it. A queue that can see the float is larger than the float behaves differently from one that can see it is not. On Sarkis's numbers the float covers the week comfortably, and that is the fact worth publishing.

TAKES AS READ

the float is 42 million and the week's demand is uncertain • the short-run trade-off between inflation and unemployment — taken as read

VERDICT 103W

The float covers the week comfortably on Sarkis's figures, and that is the fact that stops a run rather than a limit. A limit protects the money. It also announces that the board expects the money to need protecting, which is a reason to join the queue. The protection and the signal arrive together and cannot be separated. Setting a limit and publishing the float contradicts itself: if the float is sufficient, the limit is unexplained. Opening late concedes the point before anybody has asked the question, and the queue at ten to nine is already the length it is going to be.

TAKEAWAY 12W

Every protective measure is also a statement about what is being protected.

15.3 Four figures that used to be opinions

RATE ROOM · A ROOM · SEQUENCE

SCENE 24W

Verhoeven has the four things that changed this fortnight on one page. She wants them in an order the next board can act on.

GUIDE 45W

Look for what the four have in common rather than what each says. In every case a figure that had been agreed was replaced by one that was measured, with a period or a condition attached. Order them by what the next board needs first.

ASK 11W

Put the four in the order the next board needs them.

BEHIND THE BUTTON 71W

What each replaced. A basket from another decade. A money supply that double-counted the interbank deposits. A rate chosen for fairness rather than for what reserves could hold. And a multiplier nobody had computed.

Why the order is by urgency. The next board inherits the file and needs first what it cannot work without. That is the index, then the money supply figure, then the reserve arithmetic, then the requirement schedule.

TAKES AS READ

the file passes to the next board intact · a price index, and the basket it is an index of — taken as read

VERDICT 119W

The order is by what stops work rather than by what matters most. Nothing can be said about wages, output or debt without the index, because every real figure is deflated by it. The money supply comes next: it is what the price path is argued from, and its interbank correction is the part that will be forgotten first. The reserve arithmetic is needed when the rate is questioned, which is weeks away rather than days. The requirement schedule is last because it is already in force and runs itself. All four are the same move: an agreed figure replaced by a measured one, with its conditions written beside it. That is what the next board is really inheriting.

TAKEAWAY 18W

An agreed figure and a measured one can be the same number and are not the same evidence.

END OF THE DAY 14W

A measured figure with its conditions stated is worth more than an agreed one.

THE CLOSING CARD

The new mark went out on the fifteenth at 4.15 to the old, which is not the rate the board wanted and is the rate the reserves could hold. Prices rose eleven per cent in the first month and then stopped, which is what a changeover does when the money supply is capped and does not when it is not. The index Radic certified is published monthly with its basket printed beside it, so the next argument about inflation starts from a number somebody can check.

What is unfinished is the banks. Reserve requirements went from two per cent to eight in a fortnight, and three of the eleven regional banks cannot meet them without selling assets nobody wants to buy. That is next quarter, and it is somebody's problem.

You put a defensible index in front of the board before the rate was argued, so the rate was set against measured prices rather than against the fear in the room. You found the money the multiplier would have created and said so while there was time to cap it. Nine hundred thousand people were paid in a currency that held its value through the changeover because the arithmetic was done first.

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